Cognitive Modeling

Complete Lecture Notes

Week 15: Modeling in Cognitive Science II: Reinforcement Learning, Bayesian Cognition and AI

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Chapter 1

Applications in Cognitive Science

1.1 Week 15: Modeling in Cognitive Science II: Reinforcement Learning, Bayesian Cognition and AI

> Chapter 6: Examples of modeling in different areas of cognitive science · Lecture 1 h · Lab 2 h Running data: simulated choices in a two-armed bandit task with a reversal; a simulated visual–haptic cue-combination experiment.

1.1.1 Key terms

Important terms of this lesson, with their Persian equivalents (repeated even if they appeared in earlier weeks).

Table 1.1: Key terms

English	فارسی	توضیح کوتاه
simulation	شبیه‌سازی	تولید داده‌ی مصنوعی با اجرای مدل
bandit task	تکلیف bandit	انتخاب مکرر بین گزینه‌هایی با پاداش احتمالی
cue combination	ترکیب نشانه‌ها (cues)	ترکیب اطلاعات چند حس برای یک برآورد
Q-learning	Q-learning	قاعده‌ی یادگیری ارزش گزینه‌ها
softmax / inverse temperature (β)	softmax / دمای معکوس	تبدیل ارزش‌ها به احتمال انتخاب
learning rate (α)	نرخ یادگیری	اندازه‌ی به‌روزرسانی بعد از هر خطا
temperature	دما (temperature)	پارامتر نویز در انتخاب token؛ $\beta = 1/T$

English	فارسی	توضیح کوتاه
trial	کوشش (trial)	یک بار انجام تکلیف توسط شرکت‌کننده
likelihood	درست‌نمایی	احتمال داده به‌عنوان تابعی از پارامترها
fit / fitting	برازش	تنظیم پارامترها تا مدل با داده جور شود
parameter recovery	بازیابی پارامتر	آیا برازش، پارامترهای معلوم شبیه‌سازی را برمی‌گرداند
parameter	پارامتر	کمیت قابل تنظیم مدل که بین افراد یا شرایط فرق می‌کند
ideal observer	مشاهده‌گر ایده‌آل	مدل استنباط بهینه با اطلاعات موجود
cognitive model	مدل شناختی	توصیف دقیق و ریاضی یا محاسباتی از یک فرایند ذهنی
exploration / exploitation	بهره‌برداری / کاوش	انتخاب بهترین / امتحان گزینه‌های دیگر گزینه‌ی فعلی
prediction error	خطای پیش‌بینی	تفاوت پاداش گرفته‌شده و پاداش مورد انتظار
reinforcement learning (RL)	یادگیری تقویتی	یادگیری از پاداش و پیامد
noise	نویز	تغییرات تصادفی در شواهد یا در داده
estimate / estimator	برآوردگر / برآورد	روش به / مقدار به‌دست‌آمده برای پارامتر دست آوردن آن
reliability	پایایی (قابلیت اطمینان)	دقت یک نشانه؛ عکس واریانس آن
large language model (LLM)	مدل بزرگ زبانی	شبکه‌ی عصبی‌ای که کلمه‌ی بعدی متن را پیش‌بینی می‌کند
parameter trade-off	بده‌بستان پارامترها	تغییر یک پارامتر که با پارامتر دیگر جبران می‌شود
RLHF / reward model	مدل پاداش / RLHF	آموزش chatbot از روی ترجیحات انسانی
Bradley–Terry model	مدل Bradley–Terry	مدل انتخاب برای مقایسه‌های جفتی

1.1.2 Learning outcomes

By the end of this week, students will be able to:

1. Formalize reward learning with the Rescorla–Wagner / Q-learning rule and a softmax choice rule, and interpret the learning rate and inverse temperature.
2. Write the trial-by-trial likelihood of a learning model, fit it by MLE, and evaluate parameter recovery and model recovery against simpler strategies.
3. Derive the predictions of a Bayesian ideal observer for cue combination, and test whether simulated behavior is optimal.

4. Distinguish Bayesian models of cognition (a claim about the mind) from Bayesian data analysis (a method for scientists).
5. Discuss how cognitive models and AI systems inform each other, and how the evaluation tools of this course apply to AI systems proposed as models of cognition.

1.1.3 Motivation & big picture

Learning from consequences. A participant repeatedly chooses between two slot machines, one paying off more often than the other. Halfway through, the machines secretly swap. How quickly does the participant notice and adapt? How much do they explore versus exploit? One of the most successful models in all of psychology and neuroscience answers with a single equation: update your expectation by a fraction of the prediction error, the difference between what you got and what you expected. The same error-driven rule appears in classical conditioning (Rescorla–Wagner), in connectionist learning (the delta rule), in the firing of dopamine neurons, and in the reinforcement-learning algorithms behind modern AI systems. Fitting it to choice data is one of the most common uses of cognitive modeling today, in computational psychiatry for instance.

The mind as a statistician. When you see and touch an object at the same time, vision and touch both give noisy estimates of its size. A *Bayesian ideal observer* combines them by weighting each cue by its reliability, which gives an estimate more precise than either alone. Do people do this? This is a different use of the word "Bayesian" from Weeks 9–13: here Bayes' rule is a hypothesis about the mind, at Marr's computational level (Week 1).

Cognitive models $\leftrightarrow$ AI. Many ideas travel in both directions between cognitive modeling and AI: error-driven learning, neural networks, reinforcement learning, Bayesian inference. Today AI systems, including large language models, are themselves proposed as models of human cognition. The tools of this course (likelihoods, model comparison, generalization tests, recovery) are exactly what is needed to evaluate such claims.

Where is this used?

- Computational psychiatry: reinforcement-learning models are fitted to choices of patients with depression, addiction, schizophrenia or OCD, to test whether they learn differently from rewards and punishments (e.g. a lower learning rate after losses).
- Neuroscience: the prediction error of the delta rule matches the activity of dopamine neurons, one of the best-known links between a cognitive model and the brain.
- Education and games: adaptive learning systems and game design use RL ideas to model how players and learners adjust to feedback.
- Perception and robotics: optimal cue combination explains how people combine vision, touch and hearing, and the same weighting rule is used in sensor fusion.
- AI: modern reinforcement learning and the training of chatbots from human preferences (RLHF) use the same choice models (see the AI & LLMs section).

1.1.4 Lecture notes

In plain words Learning from rewards. Each option (e.g. each of two slot machines) has a "value" in the participant's head: how much reward they expect from it. After choosing an option and seeing the outcome, the value is nudged toward what actually happened: *new value = old value + learning rate × (reward − old value).* The term in brackets is the prediction error, the surprise. The learning rate α says how big the nudge is. Choices are not perfectly greedy: the softmax rule picks the better-valued option more often, and the inverse temperature β sets how strongly (low β = more random exploring, high β = almost always the best). Fitting this model means finding the α and β that make the participant's actual sequence of choices most likely. The Bayesian mind: when two senses give noisy estimates of the same thing (seeing and touching an object's size), the best guess is a weighted average that trusts the more reliable sense more. People often come close to this optimal weighting; the model predicts it with no free parameters, which makes it a strong test.

Suggested timing (60 min): reward learning 20 · fitting learning models 10 · Bayesian models of cognition 15 · cognitive models and AI 10 · course synthesis 5.

1. Reinforcement learning: the delta rule and Q-learning

Rescorla–Wagner / delta rule. After choosing option o on trial t and receiving reward $r_t \in \{0, 1\}$:

$$\delta_t = r_t - Q_t(o) \quad \text{(prediction error)}, \qquad Q_{t+1}(o) = Q_t(o) + \alpha\, \delta_t,$$

and the values of unchosen options stay the same. The learning rate $\alpha \in [0, 1]$ sets how strongly new outcomes overwrite old expectations. A high α adapts quickly but is noisy; a low α is stable but slow to notice a reversal.

Softmax choice rule. Values are turned into choice probabilities with inverse temperature $\beta \geq 0$:

$$P(\text{choose } o \text{ on trial } t) = \frac{\exp(\beta\, Q_t(o))}{\sum_{o'} \exp(\beta\, Q_t(o'))}.$$

$\beta = 0$ means random choice (pure exploration); a large β means always picking the currently best option (exploitation). For two options this is a logistic function of the value difference: $P(o_1) = 1/(1 + e^{-\beta(Q_1 - Q_2)})$.

Connections: - The update is the delta rule of connectionist networks (Verguts, 2022): a single-layer network trained by error correction. - Prediction errors δ_t resemble the phasic responses of midbrain dopamine neurons, which made RL models central to computational neuroscience. - Q-learning with temporal-difference errors is the basis of modern reinforcement learning in AI.

Reading the math

- $Q_t(o)$: the value of option o on trial t (the subscript t is the trial number).
- $\delta_t = r_t - Q_t(o)$: reward received minus reward expected. Positive = pleasant surprise, negative = disappointment.
- $Q_{t+1}(o) = Q_t(o) + \alpha\, \delta_t$. Example: $Q = 0.5$, $\alpha = 0.3$, reward 1: $\delta = 0.5$, new $Q = 0.5 + 0.15 = 0.65$. With no reward: $\delta = -0.5$, new $Q = 0.35$.

- Softmax for two options: $P(\text{option } 1) = 1/(1 + e^{-\beta(Q_1 - Q_2)})$. With $Q_1 - Q_2 = 0.25$: $\beta = 0$ gives 0.50 (random), $\beta = 3$ gives 0.68, $\beta = 20$ gives 0.99 (almost always option 1).

2. Fitting a learning model

The likelihood is sequential. Values depend on the participant's own past choices and rewards, so we run the model through the participant's actual trial sequence and multiply the probabilities it assigned to the choices actually made:

$$\ell(\alpha, \beta) = \sum_{t=1}^{T} \ln P\big(\text{choice}_t \mid Q_t(\cdot;\ \alpha, \text{history}_{<t}),\ \beta\big).$$

Practical points: - Reparameterize: $\alpha = 1/(1 + e^{-\eta_\alpha})$ and $\beta = e^{\eta_\beta}$ (Week 3), and use multi-start. - α–β trade-off: noisy choice (low β) and slow or erratic learning (extreme α) can produce similar-looking behavior, so the estimates are often correlated across fits, and the sign of the correlation depends on the design and the parameter range. A further problem is that very large β values are hard to estimate at all (lab Part C). Recovery studies at the actual design are essential (Week 7). - The design matters: reversals and enough trials are needed to estimate α; stable reward probabilities give little information about it. - Alternative strategies must be compared: e.g., *win-stay/lose-shift* (WSLS: repeat a rewarded choice, switch after no reward, with lapse rate ε), or a biased-random model. A good RL fit means little if WSLS fits just as well.

Reading the math $\ell(\alpha, \beta) = \sum_t \ln P(\text{choice}_t \mid Q_t, \beta)$: run the learning model through the participant's *own* sequence of choices and rewards, trial by trial. On each trial, compute the probability the model gave to the choice the participant actually made, take its log, and add up. The worked example (Part 1) does this for four trials: $\ln 0.5 + \ln 0.679 + \ln 0.593 + \ln 0.755 = -1.884$. The best (α, β) maximize this sum. It is Week 4's likelihood, except that each trial's probability depends on the history before it.

3. Bayesian models of cognition

Cue combination. A visual cue $x_V \sim \mathcal{N}(S, \sigma_V)$ and a haptic cue $x_H \sim \mathcal{N}(S, \sigma_H)$ both measure the true size S. With a flat prior, the posterior over S is normal with mean

$$\hat{S} = w_V x_V + w_H x_H, \qquad w_V = \frac{1/\sigma_V^2}{1/\sigma_V^2 + 1/\sigma_H^2}, \quad w_H = 1 - w_V,$$

and SD

$$\sigma_{VH} = \sqrt{\frac{\sigma_V^2\,\sigma_H^2}{\sigma_V^2 + \sigma_H^2}} \ \le\ \min(\sigma_V, \sigma_H).$$

Testable predictions (tested, e.g., by Ernst & Banks, 2002): (1) the weights should track the cues' reliabilities, which the experimenter can manipulate by degrading one cue; (2) bimodal precision should be *better* than the best single cue, by exactly the amount above. The single-cue SDs are

measured separately, so the ideal-observer predictions have no free parameters. That makes them strong, falsifiable tests.

What a Bayesian model of cognition claims. That behavior approximates optimal inference given the agent's noise, prior and goals: a computational-level account (Marr). It is silent about the algorithm: the brain might approximate Bayes with sampling, heuristics or neural population codes.

Criticisms to keep in mind: with free choices of priors, likelihoods and loss functions, almost any behavior can be called "optimal" after the fact (the flexibility problem of Week 6). Strong tests fix these components independently, as in cue combination.

Two meanings, one more time:

Table 1.2: 3. Bayesian models of cognition

	Bayesian model of cognition	Bayesian data analysis
Who does the inference?	The participant's mind (hypothetically)	The scientist
Prior represents	The participant's beliefs about the world	The scientist's knowledge about parameters
Can be wrong if...	People are not optimal	The model or priors are poorly chosen
Examples in this course	Cue combination (today)	Weeks 9–13

You can analyze a Bayesian model of cognition with maximum likelihood, and a non-Bayesian model (e.g., Q-learning) with Bayesian data analysis.

Reading the math

- $w_V = \dfrac{1/\sigma_V^2}{1/\sigma_V^2 + 1/\sigma_H^2}$: the weight of vision is its precision (1/variance) divided by the total precision. A sense twice as precise (half the SD, a quarter of the variance) gets four times the weight.
- $\sigma_{VH} = \sqrt{\sigma_V^2\sigma_H^2/(\sigma_V^2 + \sigma_H^2)}$: the combined SD, always smaller than the better single sense. Example: $\sigma_V = 1$, $\sigma_H = 2$: $w_V = 1/(1 + 1/4) = 0.8$ and $\sigma_{VH} = \sqrt{4/5} = 0.894$, about 11% better than vision alone.
- The test: measure σ_V and σ_H separately (one sense at a time), predict w_V and σ_{VH} with no free parameters, then check the two-sense data.

For an engineer

- The delta rule is the LMS update. $Q \leftarrow Q + \alpha(r - Q)$ is a one-tap LMS (or an exponentially weighted moving average with forgetting factor $1 - \alpha$): large α tracks changes quickly but is noisy, small α is smooth but slow, the familiar tracking–misadjustment trade-off.
- Softmax is the Boltzmann/Gibbs distribution; β is an inverse temperature in the physical sense. Sampling with a temperature is also how language models generate text (Week 1 AI box).

- Fitting α and β from choices is system identification of an adaptive system from its binary outputs only: you never observe Q, only the decisions it produces. That is why α and β trade off and why recovery checks are essential.
- Optimal cue combination uses exactly the inverse-variance weights of maximal-ratio combining and of sensor fusion in a Kalman filter: the combined estimate has variance $(1/\sigma_V^2 + 1/\sigma_H^2)^{-1}$. The scientific question is whether the *brain* implements this fusion.

4. Cognitive models and artificial intelligence

Table 1.3: 4. Cognitive models and artificial intelligence

Direction	Examples
Cognitive science → AI	Neural networks (inspired by neurons and learning rules); reinforcement learning (from animal learning theory); exemplar/prototype ideas in machine-learning classifiers (nearest-neighbor vs. centroid methods)
AI → cognitive science	Deep networks as models of visual cortex; RL agents as models of decision making; large language models proposed as models of language processing and of human-like reasoning
Shared mathematics	Likelihoods and loss functions; gradient-based optimization (Week 3); Bayesian inference and MCMC; cross-validation and generalization (Week 6)

Evaluating an AI system as a cognitive model uses exactly the course toolkit: - Does it predict human data *beyond* fitting it, e.g., new conditions and individual differences (generalization, Week 6)? - Does it beat simpler, interpretable cognitive models on a fair comparison (Weeks 7, 10)? - Can we tell from data whether humans and the model use the same mechanism, or only produce similar outputs (model mimicry and recovery, Week 7)? - Are its parameters or internal states interpretable as psychological constructs (Week 1)?

A model with billions of parameters that fits behavior well is not automatically an *explanation*: flexibility, identifiability and interpretability matter as much as fit.

5. Course synthesis

Across very different domains (forgetting, detection, categorization, decisions, learning, perception), the workflow has been the same: formalize → simulate → fit → evaluate → recover → interpret. The domain changes the model; the logic of modeling stays.

For the instructor

- Play the bandit game with the class. Two "machines" (two students with hidden coins or a random number app, one paying 80% of the time and one 20%), and halfway through secretly swap them. The class experiences exploration, exploitation and reversal before seeing any equation.
- Do the four Q-learning trials of the worked example on the board with a table: value before, choice, reward, prediction error, value after.
- For cue combination, use your engineering intuition openly: "combining two noisy sensors" is a natural way to present it, and students from psychology find it illuminating.
- Keep the AI discussion grounded. Section 4 can trigger long debates; use the four evaluation questions (generalization, fair comparison, mimicry, interpretability) to keep them focused.
- Typical questions.
  - *"Why is β hard to estimate?"* Once choices are almost deterministic, increasing β further hardly changes the likelihood.
  - *"Is the brain really doing Q-learning?"* It is a good description of many choice data and matches dopamine signals, but more complex models (model-based learning) also have support.
  - *"Do people always combine cues optimally?"* Often close, but not always; the lab participant over-weights touch.

1.1.5 Worked example

Part 1: four trials of Q-learning by hand

Two options; $\alpha = 0.5$, $\beta = 3$; initial values $Q(1) = Q(2) = 0.5$.

Table 1.4: Part 1: four trials of Q-learning by hand

Trial	$Q(1)$	$Q(2)$	$P(\text{choose } 1)$	Choice	Reward	δ	Update
1	0.50	0.50	0.500	1	1	$1 - 0.5 = +0.5$	$Q(1) = 0.5 + 0.5 \times 0.5 = 0.75$
2	0.75	0.50	$1/(1+e^{-3\times 0.25}) = 0.679$	1	0	$0 - 0.75 = -0.75$	$Q(1) = 0.75 - 0.375 = 0.375$
3	0.375	0.50	$1/(1 + e^{-3\times(-0.125)}) = 0.407$	2	1	$1 - 0.5 = +0.5$	$Q(2) = 0.75$
4	0.375	0.75	$1/(1 + e^{-3\times(-0.375)}) = 0.245$	2	1	$+0.25$	$Q(2) = 0.875$

Log-likelihood of these four choices: $\ln 0.5 + \ln 0.679 + \ln(1 - 0.407) + \ln(1 - 0.245) = -0.693 - 0.387 - 0.523 - 0.281 = -1.884$.

Interpretation. With $\alpha = 0.5$ a single non-reward halves the value of option 1, and the model predicts a switch, which the participant made. The likelihood rewards parameter values that predict the participant's actual choice sequence.

Part 2: an ideal observer

Visual SD $\sigma_V = 1$ mm, haptic SD $\sigma_H = 2$ mm. Weights: $w_V = 1/(1 + 1/4) = 0.8$, $w_H = 0.2$. Predicted combined SD: $\sqrt{(1 \times 4)/(1 + 4)} = 0.894$ mm, about 11% better than vision alone. If $x_V = 50$ and $x_H = 54$ mm, the optimal estimate is $0.8 \times 50 + 0.2 \times 54 = 50.8$ mm.

Interpretation. An observer who simply *averaged* the cues ($w = 0.5$) would have SD $\sqrt{0.25 \times 1 + 0.25 \times 4} = 1.118$ mm, *worse* than vision alone. Optimal weighting is what makes combining cues worthwhile.

1.1.6 Lab (2 hours)

Goals

1. Simulate a Q-learning agent in a reversal bandit task.
2. Fit (α, β) by MLE; run parameter recovery and diagnose the α–β trade-off.
3. Compare Q-learning with win-stay/lose-shift and biased-random models.
4. Test whether a simulated observer combines cues optimally.

Timing: Part A 20 min · Part B 20 min · Part C 25 min · Part D 30 min · Part E 25 min.

Part C: parameter recovery and the α–β trade-off (25 min)

Interpret: with 200 trials and one reversal, α is recovered well (r = 0.90) but β hardly at all (r = 0.17). This also explains the Part B fit ($\hat{\beta} = 3.8$ for a true 5.0). Look at the large-β region of the β panel. Why are high inverse temperatures hard to pin down? (Hint: once choices are almost deterministic, increasing β further barely changes the likelihood.) The recovered α and β are also correlated (0.31) although the true values were independent: a trade-off. What does this imply for a study that correlates β with, say, a clinical questionnaire?

Part E: an ideal observer for cue combination (25 min)

A simulated participant judges the size of objects seen and felt simultaneously. We know the single-cue noise levels from unimodal blocks. Does the participant combine the cues optimally?

Interpret: is this participant optimal? The empirical visual weight (0.56) is well below the optimal 0.80: the participant over-weights the less reliable haptic cue. As a result, their bimodal SD (1.006

mm) is no better than vision alone (1.000 mm), instead of the 0.894 mm an ideal observer would achieve. Note that the ideal-observer prediction had no free parameters: that is what makes this a strong test.

(We simulated this participant with a true weight of 0.65. The estimate 0.56 is off by about 3 standard errors, an unusually unlucky sample but a useful reminder that empirical weights also have sampling error. Exercise 4 puts bootstrap intervals around them.)

Exercises (increasing difficulty)

Do you get -1.884 for the log-likelihood?

Exercise 2 $\star$. Rerun Part C with 50 trials (no reversal) and with 400 trials (three reversals). How does recovery of α change? What does this imply for experimental design?

Exercise 3 $\star\star$ (two learning rates). Many studies find different learning rates for positive and negative prediction errors. Formulate and evaluate a model with α^+ and α^-, simulate an agent with $\alpha^+ = 0.4$, $\alpha^- = 0.1$, and test with AIC/BIC and a recovery study whether the extra parameter is identifiable at the Part A design.

Exercise 4 $\star\star$ (optimal weights). In Part E, simulate an observer with $w_V = 0.8$ (optimal) and one with $w_V = 0.5$ (averaging), 100 bimodal trials each. Use a bootstrap (Week 5) to put confidence intervals around the empirical weights and SDs. With 100 trials, can you distinguish the two observers?

Exercise 5 $\star\star\star$ (Advanced: hierarchical RL). Simulate 20 agents with individual (α_j, β_j) and only 80 trials each.

Lab check-off

Show: (1) the MLE fit and the recovery plots with your interpretation of the β panel, (2) the model-recovery table, and (3) your answer on the optimality of the cue-combination observer.

1.1.7 Connection to AI & LLMs

The choice rule behind chatbot training. Modern chatbots are fine-tuned with *reinforcement learning from human feedback* (RLHF). People see two responses and say which they prefer, a reward model is fitted to these choices, and the chatbot is then trained by reinforcement learning to maximize the learned reward (Christiano et al., 2017; Ouyang et al., 2022). The reward model is usually the Bradley–Terry model, $P(i \succ j) = 1/(1 + e^{-(r_i - r_j)})$. This is the softmax choice rule of this week with $\beta = 1$, and the Luce choice rule of the GCM. Fitting it is an MLE problem from Week 4. So human choice models sit at the heart of how today's AI systems are aligned with human preferences, and so do their problems: noisy choices, identifiability (only *differences* in r matter, so one value is fixed), and the gap between what people choose and what is actually good.

With about 300 noisy comparisons, the estimated order of responses 1 and 2 is even reversed. A reward model is a fitted cognitive model, and it needs uncertainty estimates and recovery checks like any other.

Is the reversal of responses 1 and 2 within the uncertainty? - ⋆⋆ Run a parameter recovery: how many comparisons are needed before the order of all four responses is recovered correctly in 90% of simulated datasets? - ⋆⋆ Simulate raters who prefer *longer* answers regardless of quality (add a length bonus to the true reward). What does the fitted reward model learn, and what will the chatbot then be trained to do?

Optional reading: Christiano, P. F., et al. (2017). Deep reinforcement learning from human preferences. *NeurIPS*. Ouyang, L., et al. (2022). Training language models to follow instructions with human feedback. *NeurIPS*.

1.1.8 Common mistakes & misconceptions

1. Fitting a learning model to aggregated choice proportions. The likelihood must follow each participant's own trial sequence, because values depend on their own history.
2. Interpreting α without a recovery check. α and β trade off, and α is poorly identified without reversals or enough trials.
3. Not comparing with simple strategies. Win-stay/lose-shift or perseveration can mimic learning. Show that the RL model wins a fair comparison.
4. Forgetting that the softmax uses value *differences*. Adding a constant to all Q-values changes nothing, while the scale of the values and β trade off.
5. "People are Bayesian" from a post-hoc fit. If the prior and likelihood were chosen to fit the behavior, optimality is assumed, not tested. Strong tests fix them independently (cue combination).
6. Confusing the two Bayesianisms (model of the mind vs. method of analysis).
7. "A large AI model that fits human data is a good cognitive model." Fit is not explanation. Ask about generalization, mimicry, identifiability and interpretability, as for any model.

1.1.9 Check your understanding

1. Write the delta-rule update and explain the role of the prediction error. Answer $Q \leftarrow Q+\alpha(r-Q)$. The prediction error $r - Q$ is the difference between outcome and expectation: positive errors raise the value, negative errors lower it, and the learning rate scales the change.

2. What does the inverse temperature β control? What behavior do $\beta = 0$ and $\beta \to \infty$ produce? Answer The sensitivity of choices to value differences (exploration vs. exploitation). $\beta = 0$: random choice regardless of values. $\beta \to \infty$: always choose the option with the highest current value.

3. Why must a learning model's likelihood be computed trial by trial on each participant's actual sequence? Answer The model's predictions on trial t depend on that participant's own previous choices and rewards (through the Q-values), so the probability of each observed choice can only be computed by running the model through the observed history.

4. Explain the α–β trade-off. Answer Different combinations of learning rate and choice noise can produce similar choice sequences. For example, sluggish adaptation can come from slow learning (low α) or from noisy, value-insensitive choice (low β). The estimates are therefore correlated and individually uncertain; the direction of the correlation depends on the design. In addition, high β values are weakly identified because near-deterministic choices look the same for many large β.

5. Vision has SD 2 and touch SD 2. What are the optimal weights and the predicted combined SD? Answer Equal reliabilities give $w_V = w_H = 0.5$, and the combined SD is $\sqrt{4 \times 4/8} = \sqrt{2} = 1.41$, better than either cue alone (2).

6. Give one example each of an idea that moved from cognitive science to AI and from AI to cognitive science. Answer Cognitive science $\rightarrow$ AI: error-driven learning (Rescorla–Wagner / delta rule) leading to neural-network training and reinforcement learning. AI $\rightarrow$ cognitive science: deep convolutional networks used as models of the ventral visual stream, or RL agents and language models proposed as models of human behavior.

7. A researcher shows that a large language model's responses correlate highly with human responses in a reasoning task. What further evidence would you ask for before accepting it as a cognitive model? Answer Prediction of new conditions and of individual differences (generalization); a fair comparison against simpler cognitive models; tests that distinguish the model's mechanism from alternatives that produce the same outputs (mimicry); and some correspondence between its internal states and psychological constructs.

1.1.10 Assignment

Use this week's lab time for project work and the course wrap-up.

1.1.11 Further reading

- Verguts (2022): the chapters on the delta rule and on reinforcement learning (error-driven learning and its neural basis) [verify chapters]. *(Main reading for Part 1.)*
- Farrell & Lewandowsky (2018): the chapters on neural network models and on models in neuroscience [verify chapter numbers].
- Anderson (2014): the chapters on neural networks and learning, which connect cognitive and computational neuroscience models [verify chapters].
- *Optional, outside the course list:* Ernst, M. O., & Banks, M. S. (2002). Humans integrate visual and haptic information in a statistically optimal fashion. *Nature, 415*, 429–433. Wilson, R. C.,

& Collins, A. G. E. (2019). Ten simple rules for the computational modeling of behavioral data. *eLife, 8*, e49547 (its running example is a bandit task like today's).

Preview for Week 16: final project presentations, and a closing session on good modeling practice.